```
desc supply;
```

columns 1

desc supply | Enter a SQL expression to filter results (use Ctrl+Space)

	Field	Type	Null	Key	Default	Extra
1	product_id	varchar(50)	YES		[NULL]	
2	product_name	varchar(50)	YES		[NULL]	
3	category	varchar(50)	YES		[NULL]	
4	supplier_name	varchar(50)	YES		[NULL]	
5	supplier_contact	varchar(50)	YES		[NULL]	
6	warehouse_location	varchar(50)	YES		[NULL]	
7	quantity_in_stock	varchar(50)	YES		[NULL]	
8	reorder_level	varchar(50)	YES		[NULL]	
9	unit_price	varchar(50)	YES		[NULL]	
10	last_restocked_date	varchar(50)	YES		[NULL]	
11	expiry_date	varchar(50)	YES		[NULL]	
12	batch_number	varchar(50)	YES		[NULL]	
13	status	varchar(50)	YES		[NULL]	

```
select * from supply;
```

<localhost> supply_chain | supply_chain_inventory_dirty

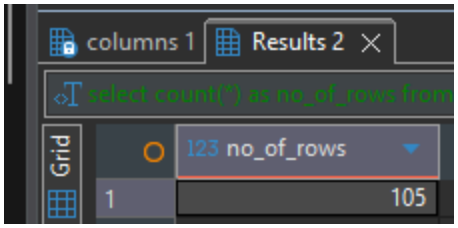
columns 1 | supply 2

Enter a SQL expression to filter results (use Ctrl+Space)

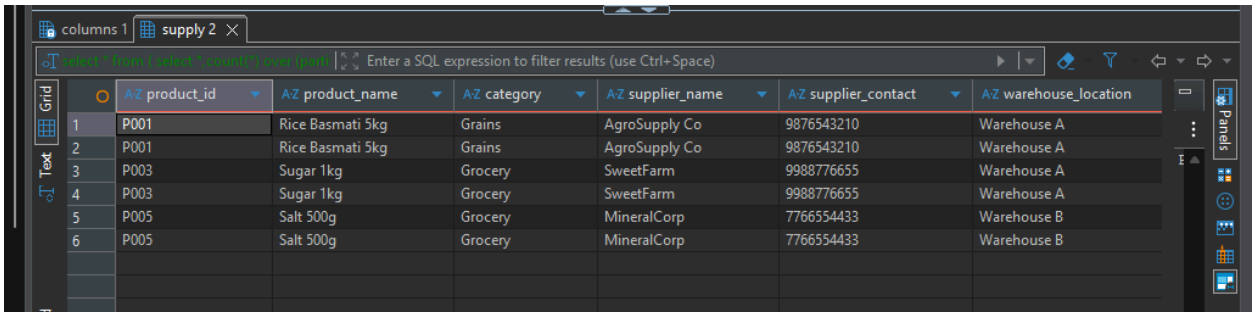
	product_id	product_name	category	supplier_name	supplier_contact	warehouse_location	quantity_in_stock
1	P001	Rice Basmati 5kg	Grains	AgroSupply Co	9876543210	Warehouse A	500
2	P002	Sunflower Oil 1L	Oils	PureOil Ltd	9123456780	Warehouse B	300
3	P003	Sugar 1kg	Grocery	SweetFarm	9988776655	Warehouse A	800
4	P004	Wheat Flour 10kg	Grains	FlourMill Inc	8877665544	Warehouse C	150
5	P005	Salt 500g	Grocery	MineralCorp	7766554433	Warehouse B	1200
6	P006	Mustard Oil 1L	Oils	GoldenOil	9811223344	Warehouse A	400
7	P007	Turmeric Powder 200g	Spices	SpiceTrade	8899001122	Warehouse B	600
8	P008	Cumin Seeds 250g	Spices	HerbsWorld	9900112233	Warehouse C	250
9	P009	Red Chilli Powder 100g	Spices	SpiceKing	9811001122	Warehouse A	450
10	P010	Coriander Powder 200g	Spices	SpiceWorld	9700990011	Warehouse B	380
11	P001	Rice Basmati 5kg	Grains	AgroSupply Co	9876543210	Warehouse A	500
12	P005	Salt 500g	Grocery	MineralCorp	7766554433	Warehouse B	1200
13	P003	Sugar 1kg	Grocery	SweetFarm	9988776655	Warehouse A	800
14	P011		Oils	PureOil Ltd	9123456780	Warehouse B	200
15	P012	Mustard Seeds 250g		SpiceTrade	8899001122	Warehouse A	600
16	P013	Cardamom 100g	Spices		7712345678	Warehouse C	400
17	P014	Cloves 50g	Spices	SpiceHub		Warehouse A	200
18	P015	Pepper Black 100g	Spices	SpiceKing	9900112233		180
19	P016	Cinnamon Sticks 50g	Spices	SpiceHub	9900112233	Warehouse A	
20	P017	Green Cardamom 50g	Spices	EliteSpice	9811223344	Warehouse B	150

Refresh | Save | Cancel | Export data | 200 | 105 | 105 row(s) fetched - 0.003s (0.001s fetch), on 2026-04-15 at 23:22:38

```
select count(*) as no_of_rows from supply;
```

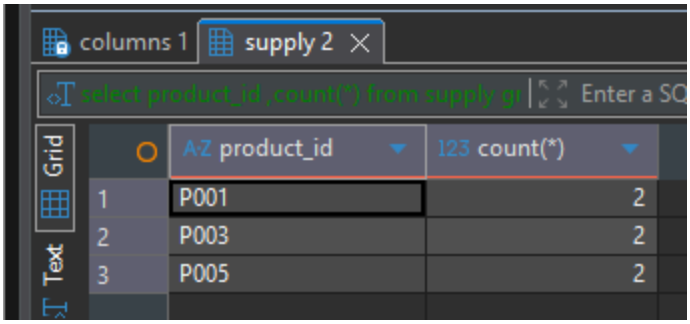



```
-- find duplicate product id values
select * from
( select *,count(*) over (partition by product_id ) as cnt
from supply
) t
where cnt > 1;
```



Or alternative

```
select product_id ,count(*)
from supply
group by product_id
having count(*) > 1;
```



columns 1 | supply 2

select product_id ,count(*) from supply gr | Enter a SQ

	product_id	count(*)
1	P001	2
2	P003	2
3	P005	2

```
-- counting number of nulls in each column
select sum(product_id is null or trim(product_id)='')  as null_product_id,
sum(product_name is null or trim(product_name)='')  as null_product_name,
sum(category is null or trim(category)='')  as null_category,
sum(supplier_name is null or trim(supplier_name)='')  as null_supplier_name,
sum(supplier_contact is null or trim(supplier_contact)='')  as null_supplier_contact,
sum(warehouse_location is null or trim(warehouse_location)='')  as null_warehouse_location,
sum(quantity_in_stock is null or trim(quantity_in_stock)='') as null_quantity_in_stock,
sum(reorder_level is null or trim(reorder_level)='')  as null_reorder_level,
sum(unit_price is null or trim(unit_price)='')  as null_unit_price,
sum(last_restocked_date is null or trim(last_restocked_date)='')  as null_last_restocked_date,
sum(expiry_date is null or trim(expiry_date)='')  as null_expiry_date,
```

```
sum(batch_number is null or trim(batch_number)='')  as
null_batch_number,
sum(status is null or trim(status)='')  as null_status
from supply;
```

null_product_id	null_product_name	null_category	null_supplier_name	null_supplier_contact	null_warehouse_location	null_quantity_in_stock	null_reorder_level	null_unit_price	null_last_r
0	4	4	4	4	4	6	6	6	

```
-- identify text in  numerical columns
SELECT product_id, product_name, quantity_in_stock,
unit_price, reorder_level
FROM supply
WHERE
  quantity_in_stock NOT REGEXP '^[0-9]+$'
OR
  unit_price NOT REGEXP '^[0-9]+$'
OR
  reorder_level NOT REGEXP '^[0-9]+$';
```

product_id	product_name	quantity_in_stock	unit_price	reorder_level
P016	Cinnamon Sticks 50g		210	55
P017	Green Cardamom 50g	150	350	
P018	Bay Leaves 50g	300		80
P019	Chickpeas 1kg	-300	95	100
P020	Red Lentils 500g	400	75	-50
P021	Groundnut Oil 2L	220	-180	80
P022	Black Pepper 200g	-100	190	40
P023	Sesame Oil 500ml	180	220	-30
P024	Soya Chunks 200g	abc	65	50
P025	Maize Flour 2kg	350	160	bad
P026	Coconut Oil 500ml	270	expensive	90
P027	Poha 1kg	two hundred	65	80
P028	Semolina 1kg	420	fifty five	100
P080	???			
P081				
P082				
P083	unknown			
P102				

```sql
-- find negative numerical columns
SELECT product_id, quantity_in_stock, reorder_level, unit_price
FROM supply
WHERE
   TRIM(quantity_in_stock) LIKE '-%'
OR  TRIM(reorder_level) LIKE '-%'
OR  TRIM(unit_price) LIKE '-%';
```

```sql
SELECT product_id, quantity_in_stock, reorder_level, unit_price
FROM supply
WHERE
   TRIM(quantity_in_stock) LIKE '-%'
OR  TRIM(reorder_level) LIKE '-%'
OR  TRIM(unit_price) LIKE '-%'
```

product_id	quantity_in_stock	reorder_level	unit_price
P019	-300	100	95
P020	400	-50	75
P021	220	80	-180
P022	-100	40	190

P023	180	-30	220

```
-- show all distinct status values
select distinct status,count(*) as cnt
from supply
group by status
order by cnt desc;
```

```
select distinct status,count(*) as cnt
from supply
group by status
order by cnt desc
```

status	cnt
Active	92
inactive	4
	3
pending	2
???	2
unknown	1
NA	1

```
-- show all distinct category values
select distinct category,count(*) as cnt
from supply
group by category
order by cnt desc;
```

```
select distinct category,count(*) as cnt
from supply
group by category
order by cnt desc
```

category	cnt
Spices	20
Grocery	17
Pulses	9
dry fruits	9
Grains	7
Oils	6
Condiments	5
	4
Beverages	4
Dairy	4
Seeds	4
cereals	3
Baking	3
Unknown	2
Protein	1
Canned Goods	1
DryFruits	1
dry-fruits	1
Natural	1
Test	1
???	1
NA	1

```sql
-- Show all distinct WAREHOUSE values
select distinct warehouse_location,count(*) as cnt
from supply
group by warehouse_location
order by cnt desc;
```

```sql
-- Show all distinct WAREHOUSE values
select distinct warehouse_location,count(*) as cnt
from supply
group by warehouse_location
order by cnt desc
```

warehouse_location	cnt
Warehouse A	35
Warehouse B	31
Warehouse C	27
	4
Warehouse X	2
wh-a	1
WH B	1
Warehouse-C	1
???	1
unknown	1
NA	1

```sql
-- Find invalid supplier contacts
SELECT product_id, supplier_name, supplier_contact
FROM supply
WHERE
  supplier_contact IS NULL
```

```
  OR TRIM(supplier_contact) = ''
  OR NOT TRIM(supplier_contact) REGEXP '^[6-9][0-9]{9}$';
```

```
SELECT product_id, supplier_name, supplier_contact
FROM supply
WHERE
   supplier_contact IS NULL
   OR TRIM(supplier_contact) = ''
   OR NOT TRIM(supplier_contact) REGEXP '^[6-9][0-9]{9}$'
```

product_id	supplier_name	supplier_contact
P014	SpiceHub	
P029	SpiceWorld	12345
P030	ElitSpice	abc987xyz
P031	SpiceHub	0
P032	SpiceHub	-9988776655
P033	HerbsCo	98765abc12
P034	SpiceKing	000
P078	TestSupplier	1234567890
P079	Unknown	0000000000
P080	???	???
P081		
P082		
P083	unknown	unknown
P101	NA	NA
P102		

```
-- Find expiry date BEFORE restock date
SELECT product_id, product_name, last_restocked_date,
expiry_date
```

```
FROM supply
WHERE
  DATE_FORMAT(
    CASE
      WHEN expiry_date LIKE '____-__-__'
        THEN STR_TO_DATE(expiry_date, '%Y-%m-%d')
      WHEN expiry_date LIKE '__-___-__'
        THEN STR_TO_DATE(expiry_date, '%d-%b-%y')
      WHEN expiry_date LIKE '__/__/____'
        THEN STR_TO_DATE(expiry_date, '%d/%m/%Y')
    END,
    '%Y-%m-%d'
  )
  <
  DATE_FORMAT(
    CASE
      WHEN last_restocked_date LIKE '____-__-__'
        THEN STR_TO_DATE(last_restocked_date,
'%Y-%m-%d')
      WHEN last_restocked_date LIKE '__-___-__'
        THEN STR_TO_DATE(last_restocked_date,
'%d-%b-%y')
      WHEN last_restocked_date LIKE '__/__/____'
        THEN STR_TO_DATE(last_restocked_date,
'%d/%m/%Y')
    END,
    '%Y-%m-%d'
  );
```

```sql
-- Find expiry date BEFORE restock date
SELECT product_id, product_name, last_restocked_date, expiry_date
FROM supply
WHERE
   DATE_FORMAT(
       CASE
           WHEN expiry_date LIKE '____-__-__'
               THEN STR_TO_DATE(expiry_date, '%Y-%m-%d')
           WHEN expiry_date LIKE '__-___-__'
               THEN STR_TO_DATE(expiry_date, '%d-%b-%y')
           WHEN expiry_date LIKE '__/__/____'
               THEN STR_TO_DATE(expiry_date, '%d/%m/%Y')
       END,
       '%Y-%m-%d'
   )
   <
   DATE_FORMAT(
       CASE
           WHEN last_restocked_date LIKE '____-__-__'
               THEN STR_TO_DATE(last_restocked_date, '%Y-%m-%d')
           WHEN last_restocked_date LIKE '__-___-__'
               THEN STR_TO_DATE(last_restocked_date, '%d-%b-%y')
           WHEN last_restocked_date LIKE '__/__/____'
               THEN STR_TO_DATE(last_restocked_date, '%d/%m/%Y')
       END,
       '%Y-%m-%d'
   )
```

product_id	product_name	last_restocked_date	expiry_date
P043	Paneer 200g	2024-03-25	2024-03-01
P044	Butter 100g	2024-03-20	2024-03-10
P045	Ghee 500ml	2024-03-18	2024-03-05
P046	Yogurt 400g	2024-03-22	2024-03-15

```sql
-- Find zero-stock products marked as Active
SELECT product_id, product_name, quantity_in_stock, status
FROM supply
WHERE TRIM(quantity_in_stock) = '0'
AND LOWER(TRIM(status)) = 'active';
```

```
SELECT product_id, product_name, quantity_in_stock, status
FROM supply
WHERE TRIM(quantity_in_stock) = '0'
AND LOWER(TRIM(status)) = 'active'
```

product_id	product_name	quantity_in_stock	status
P078	TestProduct	0	Active
P079	Dummy Item	0	Active
P084	Vanilla Extract 50ml	0	Active
P085	Baking Powder 100g	0	Active
P086	Baking Soda 200g	0	Active

```
-- -------------
-- data cleaning
-- Remove Fully Invalid / Dummy Rows
DELETE FROM supply
WHERE
  LOWER(TRIM(product_name)) IN ('testproduct','dummy
item','???','na','unknown');
```

```
-- replace null / empty
desc supply;
UPDATE supply
SET
  product_name = CASE WHEN product_name IS NULL OR
TRIM(product_name) = '' THEN 'Unknown Product' ELSE
product_name END,
  category = CASE WHEN category IS NULL OR TRIM(category)
= '' THEN 'Unknown' ELSE category END,
```

```
  supplier_name = CASE WHEN supplier_name IS NULL OR
TRIM(supplier_name) = '' THEN 'Unknown Supplier' ELSE
supplier_name END,
  warehouse_location = CASE WHEN warehouse_location IS
NULL OR TRIM(warehouse_location) = '' THEN 'Warehouse A'
ELSE warehouse_location END,
  batch_number = CASE WHEN batch_number IS NULL OR
TRIM(batch_number) = '' THEN 'Not Available' ELSE
batch_number END,
  status = CASE WHEN status IS NULL OR TRIM(status) = ''
THEN 'Active' ELSE status END;
```

```
-- Remove Duplicate product_id Records
ALTER TABLE supply
ADD COLUMN row_id INT AUTO_INCREMENT PRIMARY KEY;
DELETE t1
FROM supply t1
JOIN supply t2
ON t1.product_id = t2.product_id
AND t1.row_id > t2.row_id;
select * from supply;
```

```
-- Remove Rows with Text in Numeric Columns
DELETE FROM supply
WHERE
  TRIM(quantity_in_stock) NOT REGEXP '^-?[0-9]+$'
OR  TRIM(unit_price) NOT REGEXP '^-?[0-9]+$'
OR  TRIM(reorder_level) NOT REGEXP '^-?[0-9]+$';
```

```sql
-- Remove Rows with Negative Values
DELETE FROM supply
WHERE
   TRIM(quantity_in_stock) LIKE '-%'
OR  TRIM(reorder_level) LIKE '-%'
OR  TRIM(unit_price) LIKE '-%';
```

```sql
-- Fix Zero-Stock Products Marked as Active
UPDATE supply
SET status = 'Inactive'
WHERE TRIM(quantity_in_stock) = '0'
AND LOWER(TRIM(status)) = 'active';
```

```sql
-- Standardize STATUS Column
UPDATE supply
SET status = CONCAT(
UPPER(LEFT(LOWER(TRIM(status)), 1)),
LOWER(SUBSTR(TRIM(status), 2)));
```

```sql
-- Step 2: Remove rows with invalid status
DELETE FROM supply
WHERE status NOT IN ('Active','Inactive','Pending');
```

```sql
-- Standardize CATEGORY Column
UPDATE supply
SET category = CASE
WHEN LOWER(REPLACE(TRIM(category),'-',' ')) LIKE '%dry
fruit%' THEN 'Dry Fruits'
WHEN LOWER(TRIM(category)) IN ('cereals','cereal') THEN
'Cereals'
```

```
WHEN LOWER(TRIM(category)) IN ('grains','grain') THEN
'Grains'
WHEN LOWER(TRIM(category)) IN ('oils','oil') THEN 'Oils'
WHEN LOWER(TRIM(category)) IN ('spices','spice') THEN
'Spices'
WHEN LOWER(TRIM(category)) IN ('pulses','pulse') THEN
'Pulses'
WHEN LOWER(TRIM(category)) IN ('grocery','groceries') THEN
'Grocery'
ELSE CONCAT(UPPER(LEFT(LOWER(TRIM(category)),1)),
LOWER(SUBSTR(TRIM(category),2)))
END;
```

```
-- Standardize WAREHOUSE LOCATION
UPDATE supply
SET warehouse_location = CASE
WHEN
LOWER(REPLACE(REPLACE(TRIM(warehouse_location),'
',''),'-',''))
LIKE '%a' THEN 'Warehouse A'
WHEN
LOWER(REPLACE(REPLACE(TRIM(warehouse_location),'
',''),'-',''))
LIKE '%b' THEN 'Warehouse B'
WHEN
LOWER(REPLACE(REPLACE(TRIM(warehouse_location),'
',''),'-',''))
LIKE '%c' THEN 'Warehouse C'
ELSE NULL
END;
```

```sql
DELETE FROM supply WHERE warehouse_location IS NULL;
```

```sql
-- Remove Special Characters from Names
UPDATE supply SET
product_name =
TRIM(REPLACE(REPLACE(REPLACE(REPLACE(REPLACE(
REPLACE(REPLACE(REPLACE(REPLACE(REPLACE(product
_name,
'@',''),'#',''),'!',''),'$',''),'%',''),
'^',''),'&',''),'*',''),'?',''),'~','')),
supplier_name =
TRIM(REPLACE(REPLACE(REPLACE(REPLACE(REPLACE(
REPLACE(REPLACE(REPLACE(REPLACE(REPLACE(supplier
_name,
'@',''),'#',''),'!',''),'$',''),'%',''),
'^',''),'&',''),'*',''),'?',''),'~',''));
```

```sql
-- - Remove rows whose name still starts with a digit
DELETE FROM supply
WHERE LEFT(TRIM(product_name), 1) BETWEEN '0' AND '9';
```

```sql
--  Fix Invalid Supplier Contacts
UPDATE supply
SET supplier_contact = NULL
WHERE
  supplier_contact IS NULL
  OR TRIM(supplier_contact) = ''
  OR NOT TRIM(supplier_contact) REGEXP '^[6-9][0-9]{9}$';
```

```sql
--fill null values in supplier_contact
update supply
set supplier_contact ='9848786858'
where supplier_contact is null;
select * from supply;
```

```sql
-- Standardize Date Formats to YYYY-MM-DD
-- Convert DD-Mon-YY format (e.g. 03-Jan-24)
UPDATE supply
SET last_restocked_date =
DATE_FORMAT(STR_TO_DATE(last_restocked_date,'%d-%b-%
y'),'%Y-%m-%d')
WHERE last_restocked_date LIKE '__-___-__';
-- Convert DD/MM/YYYY format (e.g. 15/02/2024)
UPDATE supply
SET last_restocked_date =
DATE_FORMAT(STR_TO_DATE(last_restocked_date,'%d/%m/%
Y'),'%Y-%m-%d')
WHERE last_restocked_date LIKE '__/__/____';
-- NULL out invalid placeholder values
UPDATE supply
SET last_restocked_date = NULL
WHERE last_restocked_date IN
('future_date','99-99-2024','00-00-0000','NA','???')
OR last_restocked_date LIKE '99%';
```

```sql
-- --   -- -- -- -- --
-- Convert DD/MM/YYYY format (e.g. 15/02/2024)
```

```sql
UPDATE supply
SET expiry_date=
DATE_FORMAT(STR_TO_DATE(expiry_date,'%d/%m/%Y'),'%Y-
%m-%d')
WHERE expiry_date LIKE '__/__/____';
-- NULL out invalid placeholder values
UPDATE supply
SET expiry_date = NULL
WHERE expiry_date IN
('future_date','99-99-2024','00-00-0000','NA','???')
OR expiry_date LIKE '99%';
-- Convert DD-Mon-YY format (e.g. 03-Jan-24)
UPDATE supply
SET expiry_date =
DATE_FORMAT(STR_TO_DATE(expiry_date,'%d-%b-%y'),'%Y-
%m-%d')
WHERE expiry_date LIKE '__-___-__';
```

```sql
delete from supply
where expiry_date is null;
delete from supply
where last_restocked_date  is null;
```

```sql
-- Delete Rows Where Expiry Date < Restock Date
DELETE FROM supply
WHERE
  (
    CASE
      WHEN expiry_date LIKE '____-__-__'
```

```
            THEN STR_TO_DATE(expiry_date, '%Y-%m-%d')
        WHEN expiry_date LIKE '__-___-__'
            THEN STR_TO_DATE(expiry_date, '%d-%b-%y')
        WHEN expiry_date LIKE '__/__/____'
            THEN STR_TO_DATE(expiry_date, '%d/%m/%Y')
    END
  )
  <
  (
    CASE
        WHEN last_restocked_date LIKE '____-__-__'
            THEN STR_TO_DATE(last_restocked_date,
'%Y-%m-%d')
        WHEN last_restocked_date LIKE '__-___-__'
            THEN STR_TO_DATE(last_restocked_date,
'%d-%b-%y')
        WHEN last_restocked_date LIKE '__/__/____'
            THEN STR_TO_DATE(last_restocked_date,
'%d/%m/%Y')
    END
  );
```

```
-- update Rows with NULL Batch Number
update supply
set batch_number = 'Not available'
WHERE batch_number IS NULL
OR TRIM(batch_number) = '';
```

```
--  Create Clean Table
```

```sql
-- Create the clean table with constraints
CREATE TABLE clean_inventory (
product_id VARCHAR(10) PRIMARY KEY,
product_name VARCHAR(100) NOT NULL,
category VARCHAR(50) NOT NULL,
supplier_name VARCHAR(100) NOT NULL,
supplier_contact int,
warehouse_location VARCHAR(20) NOT NULL,
quantity_in_stock INT NOT NULL,
reorder_level INT NOT NULL,
unit_price int NOT NULL,
last_restocked_date DATE NOT NULL,
expiry_date DATE NOT NULL,
batch_number VARCHAR(20) NOT NULL,
status VARCHAR(20) NOT NULL
);
```

Insert only clean, validated rows

```sql
UPDATE supply
SET last_restocked_date = NULL
WHERE TRIM(last_restocked_date) = '';
UPDATE supply
SET expiry_date = NULL
WHERE TRIM(expiry_date) = '';
```

```sql
INSERT INTO clean_inventory (
  product_id,
```

```sql
  product_name,
  category,
  supplier_name,
  supplier_contact,
  warehouse_location,
  quantity_in_stock,
  reorder_level,
  unit_price,
  last_restocked_date,
  expiry_date,
  batch_number,
  status
)
SELECT
  product_id,
  TRIM(product_name),
  category,
  TRIM(supplier_name),
  CASE
    WHEN TRIM(supplier_contact) REGEXP '^[6-9][0-9]{9}$'
    THEN supplier_contact
    ELSE NULL
  END,
  warehouse_location,
  quantity_in_stock,
  reorder_level,
  unit_price,
  last_restocked_date,
  expiry_date,
  batch_number,
```

```sql
  status
FROM supply
WHERE
  product_name IS NOT NULL
  AND TRIM(product_name) != ''
  AND category IS NOT NULL
  AND supplier_name IS NOT NULL
  AND warehouse_location IN ('Warehouse A','Warehouse
B','Warehouse C')
  AND batch_number IS NOT NULL
  AND TRIM(batch_number) != ''
  AND status IN ('Active','Inactive','Pending')
  AND quantity_in_stock >= 0
  AND reorder_level > 0
```